Paleomagnetism of the Premier Mine Kimberlite

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Sixteen samples of kimberlite have been collected from deep underground in the Premier Mine at Cullinan in the Transvaal. The stability of the natural remanent magnetization of the samples has been well tested by alternating-field demagnetization and confirmed by thermal demagnetization. The magnetization is shown to be associated with magnetite, some of which is of primary origin and some of secondary origin. The magnetite of secondary origin was formed contemporaneously with or shortly after the emplacement of the kimberlite as a result of extensive hydrothermal alteration of primary minerals. It has not been possible thus far to decide between a thermoremanent and a chemical remanent origin for the observed magnetization. Previous paleomagnetic evidence has demonstrated that the kimberlite was emplaced in Precambrian times rather than in the Cretaceous, as was previously supposed. This evidence is supported by recently published geochronological work, which suggests an age of about 1750 m.y. The paleomagnetic pole at the time of emplacement of the kimberlite is located at 51.1°N, 37.5°E with $A_{95} = 7.5°$. This position is significantly different from, and lies approximately 17.5° away from, that obtained in a previous study of the post-Waterberg diabase sills, one of which cuts across the pipe. The age difference between the kimberlite emplacement and the sill's intrusion is not thought to be very great.

INTRODUCTION

In the Republic of South Africa there is a number of diamond-bearing kimberlite pipes. Economically, the most important occurrences are in the Kimberley region of the Cape Province and in an area to the northeast of Pretoria in the Transvaal. Mining operations have shown that the pipes of the Kimberley region intrude rocks of the Karroo system (Lower Carboniferous to Lower Jurassic) of South Africa [*du Toit*, 1954]. They also contain inclusions of material derived from Karroo formations long since eroded away. There is thus no doubt that the pipes of the Kimberley area are post-Karroo in age and *Williams* [1932], referring to other writers, has suggested a mid-Cretaceous age of emplacement or even later. Of the pipes in the Pretoria area only one, that of the Premier Mine, has been exploited extensively. This pipe cuts no formations that have been positively identified as postdating the Bushveld igneous complex (1950 ± 50 m.y. [*Nicolaysen et al.*, 1958]), but it does contain a vast xenolith of Waterberg material. The Waterberg system, which overlies the Bushveld complex, is certainly older than 1420 ± 70 m.y. and may be older than 1790 ± 70 m.y. [*Oosthuyzen*, 1964]. It has been eroded away at the site of the Premier Mine but occurs a few miles to the north. No inclusions have been found in the Premier Mine which can be identified as being derived from formations younger than the Waterberg system. However, because the Premier Mine is only 500 km from the Kimberley region and because kimberlite pipes are supposedly rare, it has been assumed until recently that the Premier Mine kimberlite was emplaced at the same time as the Kimberley occurrences.

A paleomagnetic study by *Jones and McElhinny* [1966] has shown that a diabase sill that cuts across the Premier Mine pipe must be correlated with post-Waterberg sills that outcrop in various parts of the Transvaal and neighboring Botswana. These sills are Precambrian in age, and it must therefore be concluded that the kimberlite of the Premier Mine was emplaced in Precambrian times. The large Waterberg xenolith indicates that the emplacement took place after the local base of the Waterberg system had been deposited.

Allsopp et al. [1967] have made radioactive age determinations *on the sill* in the Premier

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Mine which confirm the conclusion of *Jones and McElhinny* [1966]. The Rb-Sr method on mica concentrates from the sill yielded a firm minimum age of 1115 ± 15 m.y. A primary galena sample from the sill suggested an older age of intrusion of the sill in the vicinity of 1700 m.y., and galenas from the kimberlite itself indicated an age of 1750 ± 100 m.y. for the pipe itself.

Jones and McElhinny [1966] also reported paleomagnetic measurements on two samples of kimberlite obtained at such distance from the sill that their magnetizations would be unlikely to have been affected by the intrusion of the sill. Similar directions of magnetization were observed in these samples, and their mean direction was significantly different from that observed for the sill and its baked contact. This demonstration that kimberlite is amenable to paleomagnetic studies prompted the author to undertake an extensive sampling of the kimberlite in August 1965 principally to consolidate the preliminary results of *Jones and McElhinny* [1966] and to obtain a reliable paleomagnetic pole position for the time of magnetization of the kimberlite.

Sampling

The Premier Mine pipe (25°42′S, 28°32′E) is roughly elliptical in cross section at the surface and maintains this shape to a considerable depth. The majority of the samples were taken on the 890-foot level at which depth the axes of the ellipse are 840 and 410 meters. Thirteen blocks, oriented with a Brunton compass-clinometer, were obtained at approximately 30-meter intervals along 36 Tunnel on the 890-foot level except for a small section in the center of the pipe which was inaccessible. The positions of these blocks (numbers 1 through 13) are shown in Figure 1. The dolerite sill is approximately 80 meters thick and dips toward the north. The management of the mine kindly conducted a survey that established the relationship between 36 Tunnel Footwall and the top of the sill. This study showed that the sample obtained from the southerly end of the tunnel (the closest approach to the sill) was situated 115 meters above the sill, i.e. nearly one and a half sill thicknesses from the upper surface of the sill. Samples from the north end of the tunnel were at distances in excess of two sill thicknesses. It is considered unlikely that the magnetization of the above samples has been affected by the intrusion of the sill [*Jaeger*, 1957]. To check this point further, four blocks (14 through 17) were obtained, as shown in Figure 1, at distances 82, 67, 48, and 35 meters, respectively, from the upper surface of the sill.

From each block at least two cores, 2.5 cm in diameter, were drilled in the laboratory except for blocks 10 and 11 from which only one core each could be obtained. From each core a cylindrical specimen, 2.5 cm in length, was cut. For statistical purposes each block represents an independent sample.

Although the above technique of sampling is in many respects unsatisfactory, its use is often inevitable when samples are collected under-

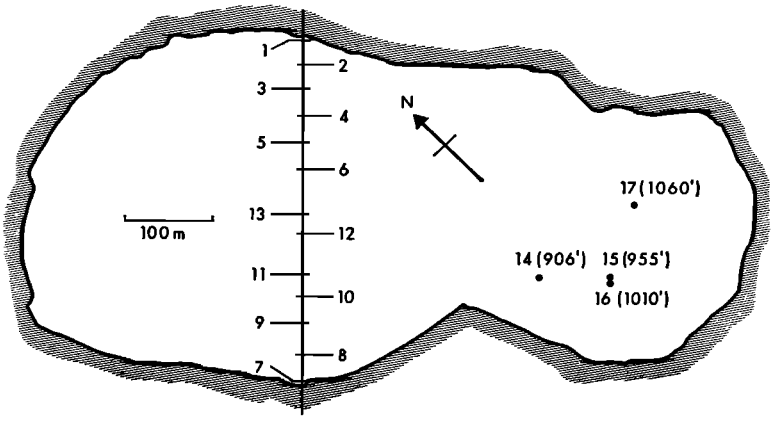


Fig. 1. Cross section of the Premier Mine pipe at the 890-foot level showing the positions from which samples were obtained along 36 Tunnel. The positions of samples 14 through 17 are also shown together with the levels from which they were taken.

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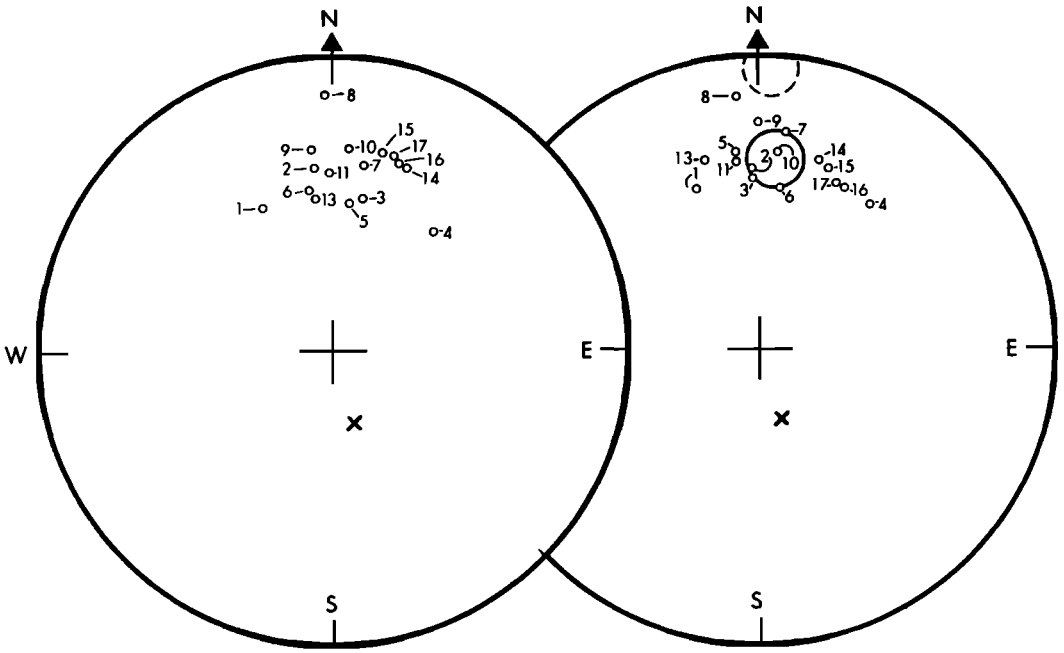


Fig. 2. Lower hemisphere stereographic projection of the directions of magnetization of the sixteen samples in the Premier Mine. (*Left*) NRM directions; (*Right*) directions of remanent magnetization after alternating-field demagnetization. Solid-line circle shows 95% confidence for the mean direction of the samples. Dashed-line circle shows 95% confidence for the mean sill direction. Open circles, south-seeking directions down. Cross, present earth's field, south-seeking down.

ground in mines. The author believes that the accuracy of orientation of individual specimens is unlikely to be better than ±3°. Nevertheless, provided that this error does not contain a systematic component, the mean results from a large collection of blocks should still be as reliable for many purposes as the results obtained using more sophisticated sampling and orienting techniques.

There is also a possibility that the magnetization of the body under consideration may cause an appreciable anomaly in the magnetic declination causing additional errors in the orientation of the blocks. In the present study a check on the magnetic declination underground indicated that the use of the declination at the surface (17° west of north) would not introduce errors greater than the ones described in the previous paragraph.

Measurements

Natural remanent magnetization. The direction of natural remanent magnetization (NRM) of each specimen was measured using a spinner magnetometer [*Gough*, 1964]. There was never more than 10° difference in the directions of NRM observed in specimens from the same sample. The magnetization of specimens from sample 12, which is a quartzite inclusion in the kimberlite, was extremely small and directions could not be determined. Figure 2 (*left*) shows the directions of NRM of all the samples except sample 12. These directions are well grouped, and the mean direction of the samples 1 through 13 lies 88° from that of the present earth's field at the mine location ($D = 343°$, $I = -63°$). This mean direction, together with the mean directions from samples 14 through 17 and from samples 1 through 17, is recorded in Table 1. Samples 14 through 17 showed very similar directions of magnetization. The specimen moments varied widely from 16.1×10^{-6} to 0.432×10^{-6} amp m^2 (16.1×10^{-3} to 0.432×10^{-3} gauss cm^3, and the corresponding intensity values are 1.31 to 3.52×10^{-2} amp m^{-1} or 1.31×10^{-3} to 3.52×10^{-5} gauss).

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TABLE 1. Mean Directions for the Premier Mine Kimberlite and for the Premier Mine Dolerite Sill

Sample Group	N	M_n D	I	R	k	B_{ac}, mwb m^{-2}	M_{ac} D	I	R	k	α_{95}
1–13	12	184	−28	11.594	27.1	30	180	−24	11.634	30.0	8.0
14–17	4	197½	−21	3.996	721	30	203	−24½	3.990	303	5.3
1–17	16	187½	−26½	15.496	29.8	30	186	−24½	15.420	25.9	7.4
Sill	10	190½	−27	8.565	6.3	10–30	183	−3	9.878	73.5	5.7

Notes

N, number of samples measured.
R, length of the resultant of N unit vectors.
D, declination of remanent magnetization.
I, inclination of remanent magnetization.
M_n, NRM.
M_{ac}, RM after demagnetization in alternating-field B_{ac}.
$k = (N - 1)/(N - R)$, estimate of precision [*Fisher*, 1953].
α_{95}, semiangle of cone of 95% confidence for mean direction.

Alternating-field demagnetization. A single specimen from each of samples 6, 8, and 17 was progressively demagnetized by the alternating-field demagnetization technique by means of the apparatus described by *McElhinny* [1966]. Normalized demagnetization curves for these specimens are shown in Figure 3. The moment of each specimen drops exponentially over most of the ranges of applied demagnetizing fields. For sample 17 no significant change in direction of magnetization was observed throughout the whole range of demagnetizing field used (0–100 mwb m^{-2}). For samples 6 and 8 little change in moment was observed in fields greater than 60 mwb m^{-2} (600 gauss). After small initial changes in direction in fields up to 15 mwb m^{-2} no further changes were observed until fields of 60 mwb m^{-2} were exceeded. At fields in excess

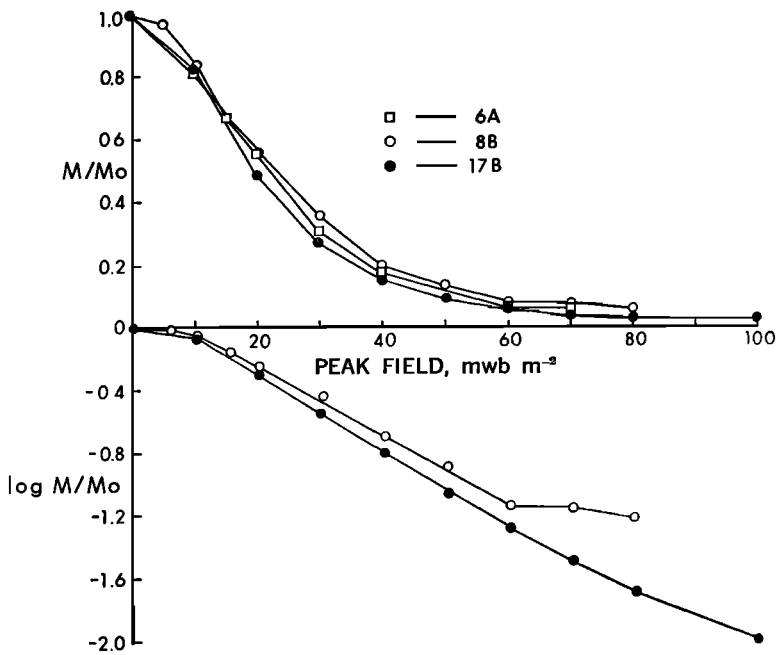


Fig. 3. Alternating-field demagnetization curves.

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of this value large erratic changes in direction were observed in both specimens. It is apparent that a stable component of magnetization was revealed in each of these specimens after the removal of a small viscous component.

All the remaining specimens used in the NRM studies were demagnetized in a field of 30 mwb m^{-2} and the results are shown in Figure 2 (*right*). The grouping of the directions noted in Figure 2 (*left*) is maintained. The mean of samples 1 through 13 now lies 92° from that of the present earth's field, and this mean together with the means from samples 14 through 17 and 1 through 17 is shown in Table 1. The change in direction after alternating-field demagnetization is not significant for any of the groupings presented, neither are the changes in the precision parameter k [*McElhinny*, 1964].

After a.f. demagnetization the mean direction of samples 14 through 17 lies 28° from that of the sill and baked contact [*Jones and McElhinny*, 1966], which is also given in Table 1. The very close grouping of samples 14 through 17, well away from the sill direction, indicates that all the samples collected were situated far enough away from the sill to not be influenced at the time of its intrusion.

The specimen moments after a.f. washing again showed a considerable range of values from 4.34×10^{-6} to 0.176×10^{-6} amp m^2. However, the ratio of the moment observed after washing in 30 mwb m^{-2} to the NRM moment was quite uniform for all these specimens (mean value 0.28 ± 0.02). This suggests that, although the individual specimens may contain greatly varying amounts of magnetic material, the nature of this material is uniform throughout the pipe.

Thermal demagnetization. Five specimens were demagnetized thermally using the technique described by *Jones et al.* [1967]. Three specimens (1C, 8D, and 14C) were demagnetized in an inert atmosphere (N_2), and the rest (8C and 16D) were treated in air. Normalized demagnetization curves are shown in Figure 4. Almost identical curves were obtained for specimens 8C and 8D, but, in general, a fairly wide range of curve types was observed. The behavior of the directions of magnetization was much more uniform. Only very small changes from the directions of NRM were observed in any of the specimens up to 550°C,

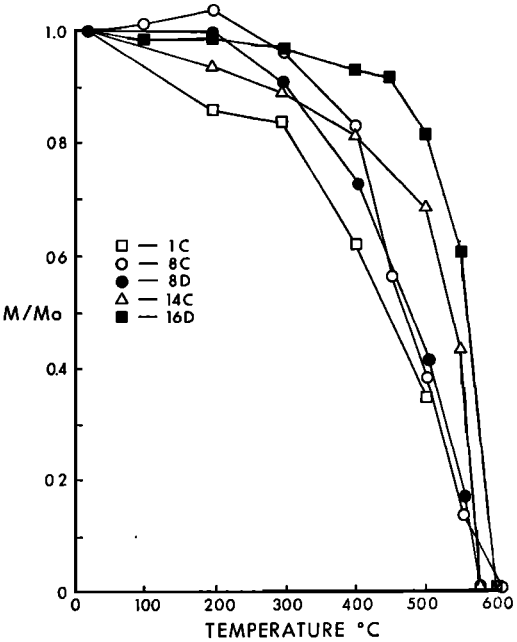


Fig. 4. Thermal demagnetization curves.

whereas large changes were recorded at 575° or 600°C. The thermal demagnetization experiments confirm the stability of the direction of magnetization observed in NRM and during alternating-field demagnetization. They also suggest that magnetite is the principal magnetic mineral responsible for the observed magnetization of the kimberlite.

Isothermal remanent magnetization. The isothermal remanent magnetization characteristics of a specimen from sample 8 were examined in the manner described by *Evans and McElhinny* [1966]. Saturation magnetization was observed for applied fields of induction greater than about 0.4 wb m^{-2}, and the increase in saturation moment observed between this value and 1.8 wb m^{-2} was less than 1%, which is the accuracy claimed for the measurement of the moments in this experiment. This experiment strengthens the suggestion previously made that magnetite is the principal magnetic material in the kimberlite.

Discussion

The discussion that follows relates to the results obtained after a.f. demagnetization unless otherwise stated. The close grouping of the directions of magnetization of the samples indi-

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cates that they acquired their magnetization after the kimberlite was emplaced and that there have been no large relative movements within the kimberlite since emplacement. The angular dispersion $\Delta = 81/k^{1/2}$ for the $N = 16$ samples is 15.9° with upper and lower limits of 21.6° and 11.7°, respectively [*Gough et al.*, 1964]. This value is therefore not significantly different from that calculated by *Creer* [1962] for the dispersion of the present earth's field from a dipolar field at low latitudes (~17° to 18°). Thus, the statistics suggest that the paleosecular variation may well be fully represented in the measurements.

The mean direction of magnetization of the sixteen samples differs from that of the Premier Mine sill and its baked contact by 22°. The circles of 95% confidence for the mean direction do not overlap (Figure 2 *right*); this suggests that the directions of magnetization for the sill and the kimberlite are significantly different. It may not be appropriate, however, to compare these directions in this manner as the time involved in the magnetization of the sill and its contact was probably very short compared with that of the kimberlite. Indeed, if secular variation has been averaged out in the kimberlite measurements as has been suggested above and if the magnetization of the sill only represents the ambient field for a very short point of time, it is possible that the sill direction represents part of the secular variation observed in the kimberlite. For a population characterized by $k = 25.9$ there is a probability of 15% that a direction will be observed 22° from the mean.

It might be more valid to compare the results of the kimberlite with those from the post-Waterberg diabases of which the Premier Mine sill is a member. A comparison of their pole positions rather than their mean directions is more appropriate, as the collection sites from the post-Waterberg diabases are scattered over a large area. Accordingly, a paleomagnetic pole position has been calculated for the $N = 16$ samples from the Premier Mine on the assumption of a geocentric axial dipole field, which is reported in Table 2. Pole positions corresponding to the mean directions of samples 1 through 13 and 14 through 17 are also given. A full statistical comparison between the polar vectors from the post-Waterberg diabases and the Premier Mine kimberlite in the manner outlined by *Jones and McElhinny* [1966] is not possible since it cannot be shown that the two populations have the same dispersions. However, the circles of 95% confidence for the two mean pole positions fail to overlap by approximately 4°. This provides a reasonable basis for asserting that the mean pole positions are significantly different and therefore that the direction of magnetization of the Premier Mine sill is significantly different from that observed in the kimberlite. Thus, the preliminary observations of *Jones and McElhinny* [1966] are confirmed.

The significant difference in the directions of magnetization in the sill and the kimberlite does not necessitate a great age difference between the times of their emplacement. Indeed, *Jones and McElhinny* [1967] in their study of the Waterberg red beds have provided evidence that both the Premier Mine kimberlite and the post-Waterberg diabases were emplaced during the period of deposition of the Waterberg system. Their conclusion finds support in the geochrono-

TABLE 2. Pole Positions for the Premier Mine Kimberlite Calculated from Group Mean Directions

Sample Group	Latitude	Longitude	δ_p	δ_m	(A_{95})	(K)
1–13	51.8°N	28.3°E	4.6°	8.6°	8.0°	30.7
14–17	45.4°N	61.7°E	3.0°	5.7°	5.2°	308
1–17	51.1°N	37.5°E	4.2°	7.9°	7.5°	24.9

Notes

δ_p and δ_m are the semiaxes of the oval of 95% confidence about the pole position and are, respectively, along and perpendicular to the paleomeridian through the pole and sampling site.

A_{95} is the semiangle of the cone of 95% confidence for a mean pole position calculated from individual sample poles.

K is the precision of site poles for this latter calculation which gives identical results to the results recorded above.

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logical work of *Allsopp et al.* [1967] outlined earlier.

Origin of Magnetization

Kimberlite is a very basic mica-olivine rock in which the olivine has been heavily changed to serpentine. *Williams* [1932] has recorded that both primary and secondary magnetite can be observed in the kimberlite. This observation has been supported by microscopic examination of some of the author's samples. Much of the secondary magnetite has resulted from the serpentinization of the olivines. This event occurred contemporaneously with the emplacement of the kimberlite or very shortly after [*Williams*, 1932; *Dawson*, 1962].

If the observed magnetization resides in the magnetite (and evidence has been provided that it does), the question arises whether it is associated with the primary magnetite (demanding a thermoremanent (TRM) origin) or with the secondary magnetite (raising the possibility of a chemical remanent (CRM) origin) or with both. It is an important question and its resolution could provide information on the temperature of emplacement of the kimberlite, a point that is still debated. If the magnetization is solely TRM, a firm *minimum* emplacement temperature of about 600°C would be implied. On the other hand, if the magnetization is solely CRM, temperatures as low as 200°C as suggested by *Davidson* [1964] (after consideration of some potassium-argon age determinations on phlogopite from Russian kimberlites) cannot be ruled out as the hydration of olivines can only take place at temperatures below 500°C [*Bowen and Tuttle*, 1949].

It is unfortunately not possible to distinguish between TRM and CRM from their alternating-field and thermal demagnetization characteristics [*Nagata*, 1961]. If the magnetization were solely TRM and the dispersion of directions comes from secular variation, a symmetry on either side of the center of the pipe (in direction of magnetization) should be apparent in samples 1 through 13. No outstanding symmetry has been recorded. However, this approach assumes a single intrusion of kimberlite followed by uniform cooling. Symmetry would be unlikely to result if the kimberlite was emplaced by multiple intrusion. This has not been definitely established in the case of the Premier Mine, but the mine geologists are able to distinguish various petrographic regions in the pipe. Further experiments to determine the precise origin of the observed magnetization are to be conducted at a later date.

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References

Allsopp, H. L., A. J. Burger, and C. van Zyl, A minimum age for the Premier kimberlite pipe yielded by biotite Rb-Sr measurements, with related galena isotopic data, *Earth Planetary Sci. Letters, 3,* 161–166, 1967.

Bowen, N. L., and O. F. Tuttle, The system MgO-SiO_2-H_2O, *Bull. Geol. Soc. Am., 60,* 439–460, 1949.

Creer, K. M., The dispersion of the geomagnetic field due to secular variation and its determination for remote times from paleomagnetic data, *J. Geophys. Res., 67,* 3461–3476, 1962.

Davidson, C. F., On diamantiferous diatremes, *Econ. Geol., 59,* 1368–1380, 1964.

Dawson, J. B., Basutoland kimberlites, *Bull. Geol. Soc. Am., 73,* 545–559, 1962.

du Toit, A. L., *The Geology of South Africa,* 3rd ed., Oliver and Boyd, Edinburgh, 1954.

Evans, M. E., and M. W. McElhinny, The paleomagnetism of the Modipe gabbro, *J. Geophys. Res., 71,* 6053–6063, 1966.

Fisher, R. A., Dispersion on a sphere, *Proc. Roy. Soc. London, A, 217,* 295–305, 1953.

Gough, D. I., A spinner magnetometer, *J. Geophys. Res., 69,* 2455–2463, 1964.

Gough, D. I., N. D. Opdyke, and M. W. McElhinny, The significance of paleomagnetic results from Africa, *J. Geophys. Res., 69,* 2509–2519, 1964.

Jaeger, J. C., The temperature in the neighborhood of a cooling intrusive sheet, *Am. J. Sci., 255,* 306–318, 1957.

Jones, D. L., and M. W. McElhinny, Paleomagnetic correlation of basic intrusions in the Precambrian of southern Africa, *J. Geophys. Res., 71,* 543–552, 1966.

Jones, D. L., and M. W. McElhinny, Stratigraphic interpretation of paleomagnetic measurements on the Waterberg red beds of South Africa, *J. Geophys. Res., 72,* 4171–4179, 1967.

Jones, D. L., M. E. R. Walford, and A. C. Gifford,

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A paleomagnetic result from the Ventersdorp lavas of South Africa, *Earth Planetary Sci. Letters, 2,* 155–158, 1967.

McElhinny, M. W., Statistical significance of the fold test in paleomagnetism, *Geophys. J., 8,* 338–340, 1964.

McElhinny, M. W., An improved method for demagnetizing rocks in alternating magnetic fields, *Geophys. J., 10,* 369–374, 1966.

Nagata, T., *Rock Magnetism,* Maruzen, Tokyo, 1961.

Nicolaysen, L. O., J. W. L. de Villiers, A. J. Burger, and F. W. E. Strelow, New measurements relating to the absolute age of the Transvaal system and of the Bushveld igneous complex, *Trans. Geol. Soc. S. Africa, 61,* 137–163, 1958.

Oosthuyzen, E. J., 'n geochronologiese studie van sekere stollingsgesteentes net ouer en jonger as die sisteem Waterberg, M. Sc. thesis, University of Pretoria, 1964.

Williams, A. F., *The Genesis of the Diamond,* vol. 1, Ernest Benn, London, 1932.

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